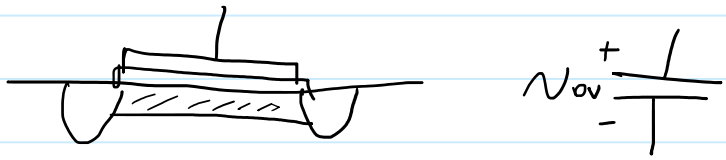
P5.2

5.2 Calculate the total charge stored in the channel of an NMOS transistor having $C_{ox} = 25\ \text{fF}/\mu\text{m}^2$, $L = 65\ \text{nm}$, and $W = 650\ \text{nm}$, and operated at $v_{OV} = 0.15\ \text{V}$ and $v_{DS} = 0\ \text{V}$.



$$C = C_{ox} \cdot W \cdot L = 25 \frac{fF}{\mu m^2} \times 0.65 \mu m \times 0.065 \mu m$$

$$C = 1.056\ fF$$

$$Q = C \cdot V = 1.056 \times 10^{-15}\ F \times 0.15\ V$$

$$= 0.15 \times 10^{-15}\ C = 0.15\ fC$$

D5.7 An n-channel MOS device in a technology for which oxide thickness is 1.4 nm, minimum channel length is 65 nm, $k_n' = 540\ \mu A/V^2$, and $V_t = 0.35$ V operates in the triode region, with small v_{DS} and with the gate–source voltage in the range 0 V to +1.0 V. If the device has the minimum channel length, what must its width be so that the minimum available resistance is 100 Ω?

$0 < V_{GS} < +1$ $\quad V_{ov} = V_{GS} - V_T \Rightarrow V_{ov} = 0.65V$

$\mu_n C_{ox} = k_n' = 540\ \mu A/V^2$ $\quad V_t = 0.35$ V

$$i_D = \mu_n C_{ox} \frac{W}{L}\left(V_{ov} - \frac{V_{ds}}{2}\right) \cdot V_{ds} \qquad L = 65\text{ nm} = 0.065\ \mu m$$

$$i_D = \underbrace{\mu_n C_{ox} \frac{W}{L}}_{k_n'} \cdot V_{ov} \cdot V_{ds} \Rightarrow i_D = g_{DS} \cdot V_{DS}$$

$$r_{DS} = \frac{V_{DS}}{i_D} = \frac{1}{g_{DS}} = \frac{1}{k_n' \frac{W}{L} V_{ov}}$$

$$100 = \frac{1}{540\mu \cdot \frac{W}{65\times10^{-9}} \cdot 0.65V} \Rightarrow W = 1852\text{ nm}$$

$$\boxed{W = 1.852\ \mu m}$$

P5.12

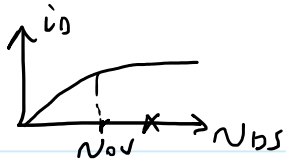
5.12 With the knowledge that $\mu_p = 0.25\mu_n$, what must be the relative width of n-channel and p-channel devices having equal channel lengths if they are to have equal drain currents when operated in the saturation mode with overdrive voltages of the same magnitude?

$$i_d = \frac{1}{2}\mu_n C_{ox}\frac{W_n}{L}v_{ov}^2 = \frac{1}{2}\mu_p C_{ox}\frac{W_p}{L}|v_{ov}|^2$$

$$\mu_n W_n = \mu_p W_p$$

$$\frac{1}{4}\frac{\mu_p}{\mu_n} = \frac{W_n}{W_p} = \frac{1}{4}$$

P5.18



5.18 A particular *n*-channel MOSFET is measured to have a drain current of 360 μA at $v_{GS} = v_{DS} = 1$ V and of 160 μA at $v_{GS} = v_{DS} = 0.8$ V. What are the values of k_n and V_t for this device?

$v_{OV} < v_{DS} \rightarrow$ sat $\qquad v_{OV} = v_{GS} - V_t < v_{DS}$

$v_{OV} > v_{DS} \rightarrow$ triode

$$i_D = \frac{1}{2}\underbrace{\mu_n C_{ox}\frac{W}{L}}_{k_n} v_{OV}^2 = \frac{1}{2}k_n(v_{GS}-V_t)^2$$

$$\frac{360\mu}{160\mu} = \frac{\frac{1}{2}k_n(1-V_t)^2}{\frac{1}{2}k_n(0.8-V_t)^2}$$

(1 − V_t: 0.6 − 0.4; 0.8 − V_t: 0.4 − 0.4)

$$\frac{36}{16}$$

$V_t = 0.4$ V

$$160\mu = \frac{1}{2}k_n(0.8-0.4)^2 \Rightarrow \frac{160\mu \cdot 2}{0.16} = k_n$$

$(0.8-0.4)^2 = 0.16$

$k_n = 2000\ \mu A/V^2$

P5.23

5.23 When the drain and gate of a MOSFET are connected together, a two-terminal device known as a "diode-connected transistor" results. Figure P5.23 shows such devices obtained from MOS transistors of both polarities. Show that

(a) the i–v relationship is given by

$$i = \frac{1}{2}k'\frac{W}{L}(v - |V_t|)^2$$

(b) the incremental resistance r for a device operating at $v = |V_t| + V_{ov}$ is given by

$$r = 1\Big/\left[\frac{\partial i}{\partial v}\right] = 1\Big/\left(k'\frac{W}{L}V_{ov}\right)$$

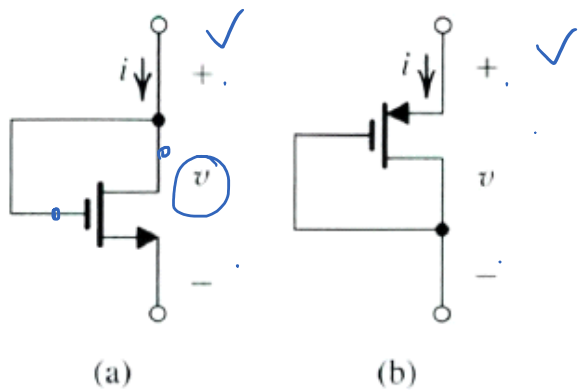


Figure P5.23

P5.27

5.27 The NMOS transistor in Fig. P5.27 has $V_t = 0.5$ V and $k_n = 2$ mA/V^2. Sketch and clearly label i_D versus v_G with v_G varying in the range 0 to +1.8 V. Give equations for the various portions of the resulting graph.

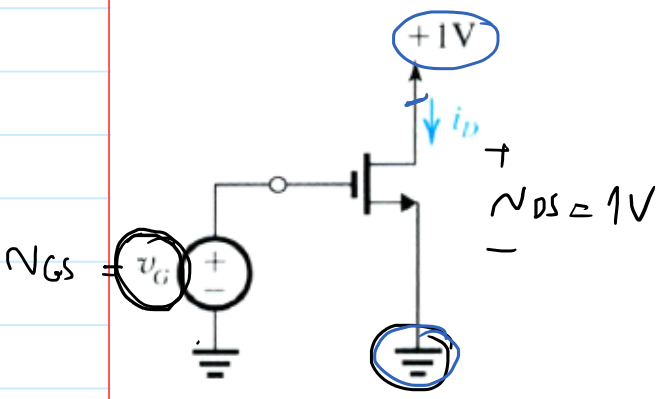
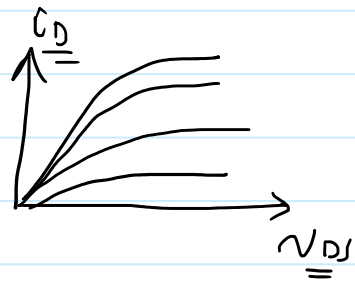


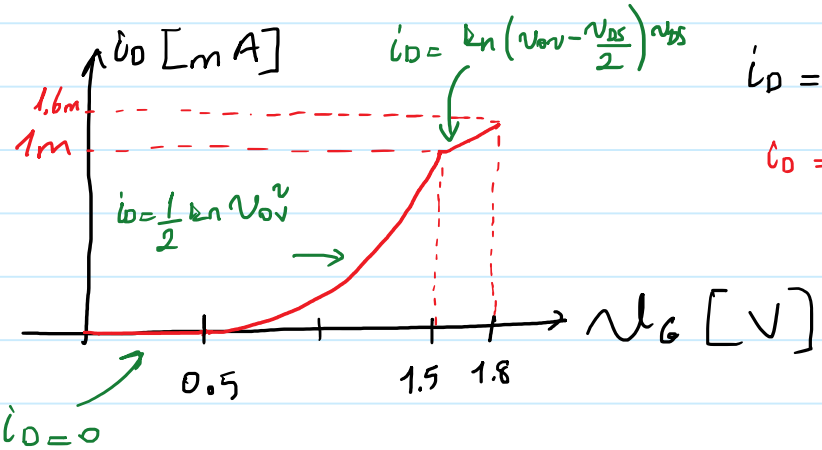
Figure P5.27



Saturation

$v_{DS} \geq v_{GS} - V_t$

$1 \geq 1.5 - 0.5$



$i_D = \frac{1}{2} k_n (v_{GS} - V_{tn})^2$

$i_D = \frac{1}{2} 2n (1.5 - 0.5)^2 = 1mA$

$$i_D = k_n \cdot \left(v_{ov} - \frac{v_{DS}}{2}\right) \cdot v_{DS} = 2n\left(v_{GS} - 0.5 - \frac{1}{2}\right) \cdot 1$$

$$i_D = 2n\,(V_G - 1) \Rightarrow @\ V_G = 1.8 \Rightarrow i_D = 2n \cdot 0.8 = 1.6m$$

P5.31

D5.31 In a particular IC design in which the standard channel length is 0.18 μm, an NMOS device of standard length with W/L of 10 operating at 100 μA is found to have an output resistance of 20 kΩ, about $\frac{1}{5}$ of that needed. What dimensional change can be made to solve the problem? What is the new device length? The new device width? The new W/L ratio? What is V_A for the standard device in this IC? The new device?

$\frac{W}{L} = 10 \quad L = 0.18\,\mu m \Rightarrow W = 1.8\,\mu m \quad I_D = 100\,\mu A$

$r_o = 20\,k\Omega \rightarrow 100\,k\Omega$

$i_D = \frac{1}{2} k_n' \frac{W}{L} v_{ov}^2 (1 + \lambda v_{DS})$ (k_n')

$R = \frac{V}{I} \qquad G = \frac{\partial I}{\partial V}$

$r_o = \frac{1}{g_{DS}} = \left[\frac{\partial i_D}{\partial v_{DS}}\right]^{-1} = \left[\frac{1}{2} k_n' \frac{W}{L} v_{ov}^2 \cdot \lambda\right]^{-1}$ (i_D')

$r_o = \frac{1}{i_D' \cdot \lambda} = \frac{V_A}{i_D'} = \frac{V_A' \cdot L}{i_D'}$

$\frac{V_A' \cdot 5L}{i_D'} \Rightarrow 5 r_o$, $L \Rightarrow 5L$, $W \Rightarrow 5W$

$L = 0.18\,\mu m \Rightarrow 5L = 0.9\,\mu m$

$W = 1.8\,\mu m \Rightarrow 5W = 9\,\mu m$

$r_o = 20\,k\Omega = \frac{V_A' \cdot 0.18\,\mu}{100\,\mu} \Rightarrow V_A = 2\,V$ (V_A)

$2 \times 10^4 \qquad 1 \times 10^{-4}$

P5.32

D5.32 For a particular n-channel MOS technology, in which the minimum channel length is 0.2 μm, the associated value of λ is 0.5 V^{-1}. If a particular device for which L is 1.0 μm operates in saturation at $v_{DS} = 1$ V with a drain current of 100 μA, what does the drain current become if v_{DS} is raised to 2 V? What percentage change does this represent? What can be done to reduce the percentage by a factor of 2?

10^{-7} ↓ 0.2μm

$L_{min} = 0.2\mu m$ $\quad \lambda = \frac{1}{2} V^{-1} = \frac{1}{V_A}$ $\quad V_A = 2V = V_A' \cdot L$

$v_{DS} = 1V$, $L = 1\mu m$, $V_A = 10V$, $\lambda = 0.1\,V^{-1}$

$i_D = \frac{1}{2} \cdot k_n' \frac{W}{L} v_{ov}^2 (1 + 0.1 v_{DS}) = 100\mu A$

$i_D' \times 1.1 = 100\mu A$ $\quad i_D' = \frac{100}{1.1}\mu A$

$\Rightarrow v_{DS} = 2V \Rightarrow i_D = ?$

$i_D = \frac{100\mu A}{1.1} \cdot (1 + 0.1 \times 2) = 109\mu A$

1.2

9% increase in i_D

$\Rightarrow v_{DS} = 2V$ $\quad i_D = 104.5\mu A$

$i_D = \frac{100\mu}{1.1} \cdot (1 + \lambda \cdot 2) = 104.5\mu$

$1.15 \Rightarrow \lambda = 0.075$

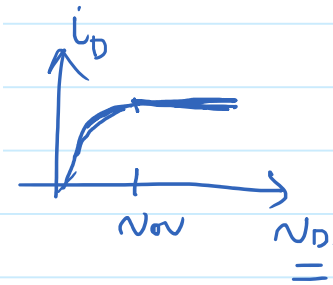
$\lambda = \frac{1}{V_A} = \frac{1}{V_A' \cdot L} = 0.075$ $\quad L = 1.33\mu m$

$10^{-7} \cdot L$

$W + \%33W$

P5.44

5.44 The NMOS transistor in the circuit of Fig. P5.43 has $V_t = 0.4$ V and $k_n = 4$ mA/V^2. The voltages at the source and the drain are measured and found to be -0.55 V and $+0.1$ V, respectively. What current I_D is flowing, and what must the values of R_D and R_S be? What is the largest value for R_D for which I_D remains unchanged?



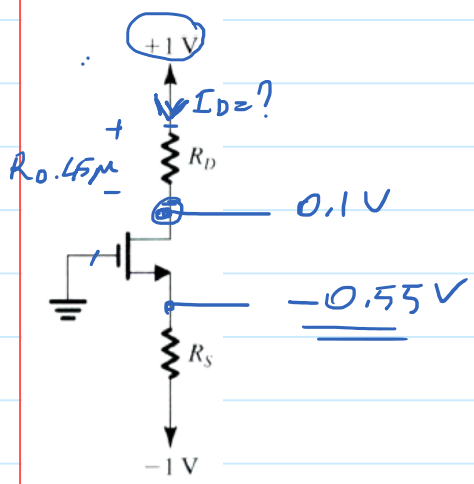


Figure P5.43

R_D 0.45μ

$v_{OV} < v_{DS}$

$v_{GS} - V_t < v_{DS}$

$0.55 - 0.4 < 0.1 - (-0.55)$

$0.15 < 0.65$

$i_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L} v_{OV}^2$

$= \frac{1}{2} k_n v_{OV}^2$

$i_D = \frac{1}{2} \cdot 4m \cdot (0.15)^2 = 45\mu A$

$R_D = \frac{1 - 0.1}{45\mu A} = 20k\Omega$

$R_S = \frac{-0.55 - (-1)}{45\mu A} = \frac{0.45}{45\mu A} = 10k$

$0.15 < v_D - v_S = 1 - R_D \cdot 45\mu - (-0.55)$

$0.15 < 1.55 - R_D \cdot 45\mu \Rightarrow R_D \cdot 45_\mu < 1.4$

$R_D < 31k\Omega$

P5.47

D5.47 It is required to operate the transistor in the circuit of Fig. P5.46 at the edge of saturation with $I_D = 50\,\mu A$. If $V_t = 0.4$ V, find the required value of R_D.

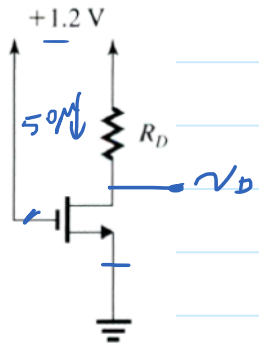


Figure P5.46

$v_{OV} \leq v_{DS}$

$v_{OV} = v_{DS}$ $I_D = 50\mu A$

$v_{GS} - V_t = 1.2 - 0.4 = 0.8$

$0.8 = v_D = 1.2 - R_D \cdot 50\mu$

$R_D = \frac{0.4}{50\mu} = 8k\Omega$

P5.55

5.55 For each of the circuits in Fig. P5.55, find the labeled node voltages. For all transistors, $k_n'(W/L) = 2$ mA/V^2, $V_t = 0.5$ V, and $\lambda = 0$.

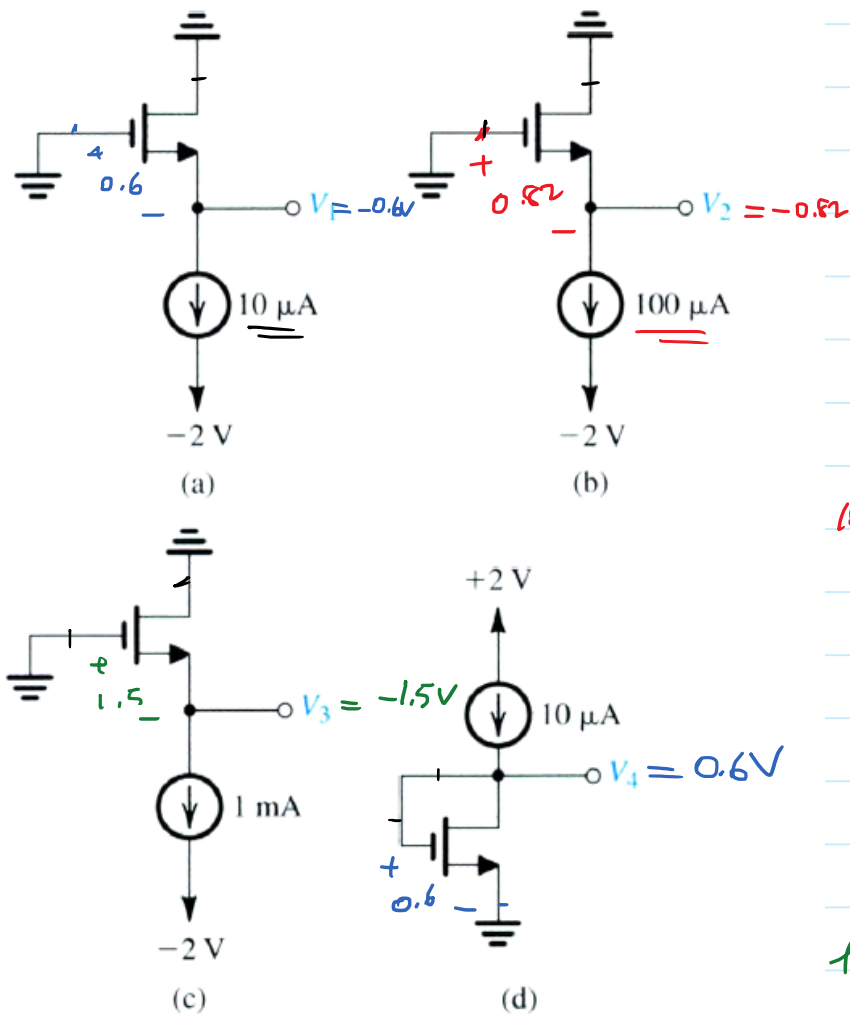


Figure P5.55

$V_{GS} - V_t < V_{DS}$

$i_{D1} = 10\mu = \frac{1}{2} 2m \cdot V_{ov}^2$

$V_{ov}^2 = 0.01 \Rightarrow V_{ov} = 0.1$

$V_{GS} - V_t = 0.1$

$V_{GS} - 0.5 = 0.1 \quad V_{GS} = 0.6$

$100\mu = \frac{1}{2} \cdot 2m \cdot V_{ov}^2$

$V_{ov}^2 = 0.1 \Rightarrow V_{ov} \simeq 0.32V$

$V_{GS} - V_t = 0.32$

$V_{GS} - 0.5 = 0.32$

$V_{GS} = 0.82$

$1m = \frac{1}{2} 2m V_{ov}^2$

$V_{ov} = 1$

$V_{GS} - V_t = 1$

$V_{GS} - 0.5 = 1 \quad V_{GS} = 1.5$

$10\mu A = \frac{1}{2} 2m V_{ov}^2$

$V_{ov}^2 = 0.01 \Rightarrow V_{ov} = 0.1V$

$V_{GS} - V_t = 0.1 \Rightarrow V_{GS} = 0.6V$

P5.56

***5.56** For each of the circuits shown in Fig. P5.56, find the labeled node voltages. The NMOS transistors have $V_t = 0.4$ V and $k_n'(W/L) = 5$ mA/V^2.

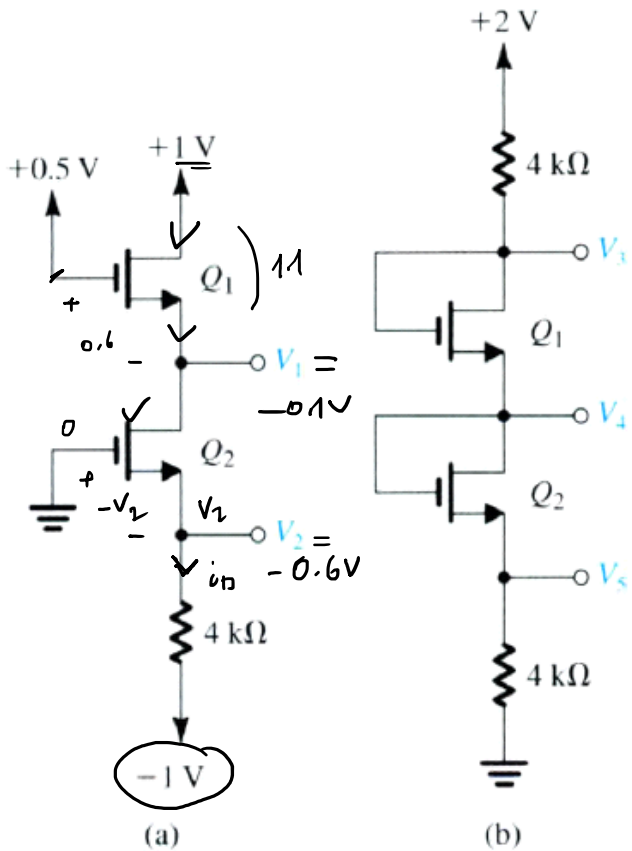


Figure P5.56

$$i_D = \frac{V_2 + 1}{4k}$$

$$i_D = \frac{1}{2} 5m (-V_2 - 0.4)^2$$

$$\frac{V_2+1}{4k} = \frac{1}{2} 5m (-V_2 - 0.4)^2$$

$$V_2 = -0.6, \; -0.1$$

$$i_D = \frac{-0.6 + 1}{4k} = 0.1m$$

$$i_D = 0.1m = \frac{1}{2} 5m (0.5 - V_1 - 0.4)^2$$

$$V_{ov} = 0.2V$$

$$0.1 - V_1 = 0.2$$

$$V_1 = -0.1\,V$$

P5.56

***5.56** For each of the circuits shown in Fig. P5.56, find the labeled node voltages. The NMOS transistors have $V_t = 0.4$ V and $k_n'(W/L) = 5$ mA/V^2.

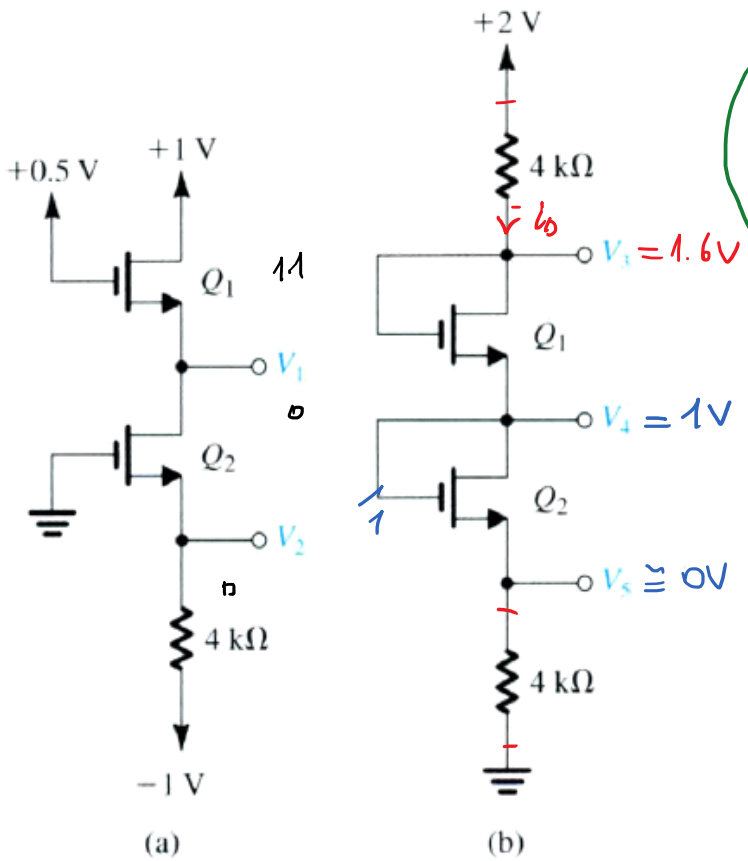


Figure P5.56

$i_D = \frac{2 - V_3}{4k}$

$i_D = \frac{1}{2} 5m (V_3 - V_4 - 0.4)^2$

$i_D = \frac{1}{2} 5m (V_4 - V_5 - 0.4)^2$

$i_D = \frac{V_5 - 0}{4k}$

$V_3 - V_4 = V_4 - V_5$

$\frac{2 - V_3}{4k} = \frac{V_5}{4k}$

$2 - V_3 = V_5$

$V_3 - V_4 = V_4 - 2 + V_3$

$2 = 2V_4$

$V_4 = 1$V

$\frac{1}{2} \cdot 5m \cdot (1 - V_5 - 0.4)^2 = \frac{V_5}{4k}$

$V_5 \cong 0.03 \simeq 0$ $V_5 \simeq 1.6$

$\frac{1}{2} 5m (V_3 - 1 - 0.4)^2 = \frac{2 - V_3}{4}$

$V_3 = 1$V $V_3 \simeq 1.6$ V